

Unit 10: Nonhomogeneous Equations and Forced Response

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb through undetermined coefficients (exponential, polynomial, sinusoidal, and the two resonance traps), the annihilator method, variation of parameters for unfriendly forcing, the transient/steady-state split, and a capstone on the impulse response and Duhamel's convolution. Solutions follow in Part III.

1. **(Undetermined coefficients — exponential)** Solve $y'' - y = e^{2x}$. Give y_h , the trial, and the general solution.
2. **(Undetermined coefficients — polynomial)** Solve $y'' - y' - 2y = 4x^2$. Use a full quadratic trial.
3. **(Undetermined coefficients — sinusoid)** Solve $y'' + 4y = \sin x$. Explain why no factor of x is needed here.
4. **(Resonance, trig)** Solve $y'' + y = \cos x$. Identify the overlap with the homogeneous solution and use the corrected resonant trial.
5. **(Resonance, exponential)** Solve $y'' - 3y' + 2y = e^x$. Show why the guess gains a factor of x .
6. **(Annihilator method)** For $y'' - y' - 2y = 2e^{3x} + x$, write the annihilator of the forcing, deduce the trial form, and solve for y_p .
7. **(Variation of parameters)** Solve $y'' + y = \sec x$, where undetermined coefficients cannot be applied.
8. **(Variation of parameters, repeated root)** Find a particular solution of $y'' - 2y' + y = \frac{e^x}{x}$ for $x > 0$.
9. **(Transient and steady-state IVP)** Solve $y'' + 4y' + 3y = 6$, $y(0) = 0$, $y'(0) = 0$. Identify the transient and the steady-state pieces.
10. **(Impulse response and Duhamel, capstone)** Find the impulse response $w(t)$ of $y'' + 3y' + 2y = f(t)$ (rest initial conditions), then write Duhamel's convolution for a general input f .

Part III — Complete Solutions

Solution 1. The homogeneous equation $y'' - y = 0$ has roots $r = \pm 1$, so $y_h = c_1 e^x + c_2 e^{-x}$. Since e^{2x} does not overlap y_h , try $y_p = Ae^{2x}$, so $y_p'' = 4Ae^{2x}$:

$$4Ae^{2x} - Ae^{2x} = 3Ae^{2x} = e^{2x} \Rightarrow A = \frac{1}{3}.$$

$$y = c_1 e^x + c_2 e^{-x} + \frac{1}{3} e^{2x}.$$

Solution 2. Roots of $r^2 - r - 2 = (r - 2)(r + 1) = 0$ give $y_h = c_1 e^{2x} + c_2 e^{-x}$. Use the complete quadratic trial $y_p = Ax^2 + Bx + C$, with $y_p' = 2Ax + B$, $y_p'' = 2A$:

$$2A - (2Ax + B) - 2(Ax^2 + Bx + C) = -2Ax^2 + (-2A - 2B)x + (2A - B - 2C).$$

Matching $-2Ax^2 + \dots = 4x^2$ term by term:

$$-2A = 4 \Rightarrow A = -2; \quad -2A - 2B = 0 \Rightarrow B = 2; \quad 2A - B - 2C = 0 \Rightarrow C = -3.$$

$$y = c_1 e^{2x} + c_2 e^{-x} - 2x^2 + 2x - 3.$$

Solution 3. Here $y'' + 4y = 0$ has roots $\pm 2i$, so $y_h = c_1 \cos 2x + c_2 \sin 2x$. The forcing $\sin x$ has frequency $1 \neq 2$, so there is *no* overlap and no factor of x is needed. Try $y_p = A \cos x + B \sin x$; then $y_p'' = -A \cos x - B \sin x$:

$$(-A + 4A) \cos x + (-B + 4B) \sin x = 3A \cos x + 3B \sin x = \sin x.$$

So $A = 0$, $B = \frac{1}{3}$, giving

$$y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{3} \sin x.$$

Solution 4. For $y'' + y = 0$ the roots are $\pm i$, so $y_h = c_1 \cos x + c_2 \sin x$. The natural guess $A \cos x + B \sin x$ is y_h — a resonance. Multiply by x : $y_p = x(A \cos x + B \sin x)$. Differentiating, $y_p'' = -x(A \cos x + B \sin x) + 2(-A \sin x + B \cos x)$, so

$$y_p'' + y_p = 2(-A \sin x + B \cos x) = \cos x \Rightarrow B = \frac{1}{2}, \quad A = 0.$$

$$y = c_1 \cos x + c_2 \sin x + \frac{x}{2} \sin x.$$

Solution 5. Roots of $r^2 - 3r + 2 = (r - 1)(r - 2) = 0$ give $y_h = c_1 e^x + c_2 e^{2x}$. The forcing e^x overlaps the e^x solution, so the bare guess Ae^x fails; multiply by x : $y_p = Axe^x$. Then $y_p' = Ae^x(1 + x)$ and $y_p'' = Ae^x(2 + x)$, so

$$Ae^x[(2 + x) - 3(1 + x) + 2x] = Ae^x(-1) = e^x \Rightarrow A = -1.$$

$$y = c_1 e^x + c_2 e^{2x} - xe^x.$$

Solution 6. The operator is $L(D) = D^2 - D - 2 = (D - 2)(D + 1)$, so $y_h = c_1 e^{2x} + c_2 e^{-x}$. The forcing $2e^{3x} + x$ is annihilated by

$$A(D) = (D - 3)D^2,$$

since $(D - 3)$ kills e^{3x} and D^2 kills the linear term x . Neither e^{3x} nor $\{1, x\}$ overlaps y_h , so the trial is $y_p = Ae^{3x} + Bx + C$. Substitute the exponential piece Ae^{3x} ($y'' - y' - 2y = 9A - 3A - 2A = 4A$):

$$4Ae^{3x} = 2e^{3x} \Rightarrow A = \frac{1}{2}.$$

For the polynomial piece $Bx + C$ ($y'' = 0$, $y' = B$):

$$\begin{aligned} -B - 2(Bx + C) &= -2Bx + (-B - 2C) = x \Rightarrow B = -\frac{1}{2}, C = \frac{1}{4}. \\ y &= c_1 e^{2x} + c_2 e^{-x} + \frac{1}{2} e^{3x} - \frac{1}{2} x + \frac{1}{4}. \end{aligned}$$

Solution 7. The equation is already in standard form with $y_1 = \cos x$, $y_2 = \sin x$, and Wronskian $W = \cos x \cos x - \sin x(-\sin x) = 1$. With $g = \sec x$,

$$\begin{aligned} u_1' &= -\frac{y_2 g}{W} = -\sin x \sec x = -\tan x \Rightarrow u_1 = \ln |\cos x|, \\ u_2' &= \frac{y_1 g}{W} = \cos x \sec x = 1 \Rightarrow u_2 = x. \end{aligned}$$

Hence $y_p = u_1 y_1 + u_2 y_2$:

$$y = c_1 \cos x + c_2 \sin x + \cos x \ln |\cos x| + x \sin x.$$

Solution 8. The repeated root $r = 1$ of $(r - 1)^2$ gives $y_1 = e^x$, $y_2 = x e^x$, with $W = e^x(e^x + x e^x) - x e^x \cdot e^x = e^{2x}$. In standard form $g = e^x/x$, so

$$u_1' = -\frac{x e^x \cdot (e^x/x)}{e^{2x}} = -1 \Rightarrow u_1 = -x, \quad u_2' = \frac{e^x \cdot (e^x/x)}{e^{2x}} = \frac{1}{x} \Rightarrow u_2 = \ln x.$$

Then $y_p = -x e^x + x e^x \ln x$; the term $-x e^x$ already lives in y_h , so

$$y = (c_1 + c_2 x) e^x + x e^x \ln x.$$

Solution 9. Roots of $r^2 + 4r + 3 = (r + 1)(r + 3) = 0$ are $r = -1, -3$, both negative, so $y_h = c_1 e^{-t} + c_2 e^{-3t}$ is the **transient**. A constant forcing calls for $y_p = K$ with $3K = 6$, so the **steady-state** is $y_p = 2$. Apply $y(0) = 0$, $y'(0) = 0$ to $y = c_1 e^{-t} + c_2 e^{-3t} + 2$:

$$\begin{aligned} c_1 + c_2 + 2 &= 0, & -c_1 - 3c_2 &= 0 \Rightarrow c_1 = -3c_2. \\ -3c_2 + c_2 + 2 &= 0 \Rightarrow c_2 = 1, & c_1 &= -3, & y &= -3e^{-t} + e^{-3t} + 2. \end{aligned}$$

As $t \rightarrow \infty$ the transient $-3e^{-t} + e^{-3t} \rightarrow 0$, leaving the steady-state value 2.

Solution (Capstone). The impulse response solves $w'' + 3w' + 2w = \delta(t)$ from rest, which for a second-order system means $w(0) = 0$, $w'(0) = 1$. Roots of $r^2 + 3r + 2 = (r + 1)(r + 2)$ are $r = -1, -2$, so $w = A e^{-t} + B e^{-2t}$:

$$\begin{aligned} w(0) &= A + B = 0, & w'(0) &= -A - 2B = 1 \Rightarrow B = -1, A = 1. \\ w(t) &= e^{-t} - e^{-2t}, & t &\geq 0. \end{aligned}$$

By Duhamel's principle the forced response from rest to any input f is

$$y_p(t) = \int_0^t w(t - \tau) f(\tau) d\tau = \int_0^t \left(e^{-(t-\tau)} - e^{-2(t-\tau)} \right) f(\tau) d\tau.$$

Unit 10 Key Takeaway

The shape of the forcing chooses the tool: **undetermined coefficients** for exponentials, polynomials, and sinusoids — with the **×x resonance fix** whenever a trial term solves $L[y] = 0$ — **variation of parameters** for $\sec x$, e^x/x , and other unfriendly inputs, and the **annihilator** to derive the trial form. In a stable system the **transient** decays to the **steady-state**, and the **impulse response** $w(t) = e^{-t} - e^{-2t}$ feeds **Duhamel's convolution** to resolve every input at once.